

Conjecture 00000000539 is false:
the h-vector $(1, 4, 6, 4, 1)$ of $(x_0^2, x_1^2, x_2^2, x_3^2)$

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1 The conjecture

Definition: The h -polynomial $h(t)$ is the encoding generating function of the Hilbert function of a graded ideal. *Conjecture:* The h -vector of a quadratic monomial ideal has no three adjacent strictly decreasing terms; this combinatorial positivity is induced by the shellability of the complex.

The conjecture is a conjunction. Its first clause is a universal statement about all quadratic monomial ideals. We refute that clause, so the conjunction is false. The second clause (the "explanation" by shellability) is not separately formalized.

2 Reading and definitions

Let K be any field and $S = K[x_0, \dots, x_{n-1}]$ with the standard grading ($\deg x_i = 1$). For a set G of exponent vectors, the *monomial ideal* generated by G is $I_G = (x^g : g \in G)$. A *quadratic monomial ideal* is an ideal I_G with every $g \in G$ of total degree 2. Squares x_i^2 are allowed; nothing in the text requires squarefree generators.

Definition 1 (Hilbert function, Hilbert series, h -polynomial). For a homogeneous ideal I , $(S/I)_j$ is the image of the degree- j homogeneous polynomials S_j under $S \rightarrow S/I$, and $\text{HF}_{S/I}(j) = \dim_K(S/I)_j$. The Hilbert series is $\text{HS}_{S/I}(t) = \sum_j \text{HF}_{S/I}(j) t^j \in \mathbb{Z}[[t]]$. A polynomial $h \in \mathbb{Z}[t]$ is the h -polynomial of S/I if $\text{HS}_{S/I}(t) (1-t)^d = h(t)$ for some $d \geq 0$ and $h(1) \neq 0$. Its coefficient list $(h_0, \dots, h_s)$, $s = \deg h$, is the h -vector.

Conventions. (1) Reading A (standard): "the Hilbert function of a graded ideal I " is that of the quotient S/I . (2) Reading B (literal): it is the Hilbert function $\text{HF}_I(j) = \dim_K I_j$, $I_j = I \cap S_j$, of the graded module I , with h -polynomial defined by the same rule from HS_I . (3) "Three adjacent strictly decreasing terms" means an index i with $i+2 \leq \deg h$ and $h_i > h_{i+1} > h_{i+2}$. Coefficients are integers. (4) $\mathbb{N}$ contains 0.

3 The counterexample

Let $n = 4$ and $I = (x_0^2, x_1^2, x_2^2, x_3^2)$, a quadratic monomial ideal.

Lemma 2 (standard monomials). *For a monomial ideal I_G in finitely many variables, $\dim_K(S/I_G)_j$ equals the number of exponent vectors m of degree j such that no $g \in G$ satisfies $g \leq m$ coordinate-wise.*

Proof. A polynomial lies in I_G if and only if every monomial in its support is divisible by some x^g , $g \in G$ (Mathlib: `MvPolynomial.mem_ideal_span_monomial_image`). Hence a nonzero combination of standard monomials (those divisible by no x^g) is never in I_G , so their images in S/I_G are linearly independent. Every non-standard monomial maps to 0, and S_j is spanned by the degree- j monomials. So the images of the degree- j standard monomials form a basis of $(S/I_G)_j$. $\square$

Theorem 3 (Reading A). $\text{HF}_{S/I}(j) = \binom{4}{j}$, so $\text{HS}_{S/I}(t) = (1+t)^4$. The h -polynomial is uniquely $(1+t)^4$, with h -vector $(1, 4, 6, 4, 1)$, and $6 > 4 > 1$ are three adjacent strictly decreasing terms.

Proof. For $G = \{2e_0, \dots, 2e_3\}$, m is standard iff every $m_i \leq 1$. Such m of degree j correspond bijectively to their supports, the j -subsets of $\{0, 1, 2, 3\}$, so Lemma 2 gives $\binom{4}{j}$. Then $(1+t)^4 \cdot (1-t)^0$ is a valid h -polynomial, $h(1) = 16 \neq 0$. If $\text{HS}(t)(1-t)^d = h$ with $h(1) \neq 0$, then $h = (1+t)^4(1-t)^d$ as polynomials and $h(1) = 16 \cdot 0^d$, so $d = 0$ and $h = (1+t)^4$. $\square$

Theorem 4 (Reading B). $\text{HF}_I(j) = \binom{j+3}{3} - \binom{4}{j}$ and $\text{HS}_I(t)(1-t)^4 = 1 - (1+t)^4(1-t)^4 = 4t^2 - 6t^4 + 4t^6 - t^8 =: h_I(t)$, with $h_I(1) = 1 \neq 0$. Every h -polynomial of I equals h_I ; its h -vector $(0, 0, 4, 0, -6, 0, 4, 0, -1)$ has $4 > 0 > -6$ in positions 2, 3, 4.

Proof. Rank-nullity for $S_j \rightarrow (S/I)_j$ (kernel I_j) gives $\dim I_j + \dim(S/I)_j = \dim S_j$, and $\dim S_j = \binom{j+3}{3}$ (number of degree- j exponent vectors in four variables). With $\sum_j \binom{j+3}{3} t^j \cdot (1-t)^4 = 1$ the identity follows. Uniqueness: if $F(1-t)^a = p$ and $F(1-t)^b = q$ with $a \leq b$ and $q(1) \neq 0$, then $q = p(1-t)^{b-a}$, forcing $a = b$ and $p = q$. $\square$

Remark (not formalized). The squarefree polarization $J = (x_0y_0, x_1y_1, x_2y_2, x_3y_3) \subset K[x_i, y_i]$ (edge ideal of a perfect matching on 8 vertices; its Stanley–Reisner complex is the boundary of the 4-dimensional cross-polytope) has the same h -vector $(1, 4, 6, 4, 1)$: its degree- j standard monomials correspond to pairs (subset s of blocks with a y -power, arbitrary exponent vector b of degree $j - |s|$), so $\text{HS}_{S/J} = (1+t)^4/(1-t)^4$. This is only a paper argument; the Lean development covers the ideal I above.

4 Lean formalization

File `lean/Conjecture539/Basic.lean` (Lean 4, Mathlib v4.33.1, only `import Mathlib`).

- Objects: `monomialIdeal`, `IsQuadraticMonomialIdeal`, `gradedPiece` (image of homogeneous submodule in S/I), `hilbertFunction` (`finrank`), `hilbertSeries`, `IsHPolynomial`, `HasTripleDescent`; for reading B `idealPiece`, `idealHilbertFunction`, `idealHilbertSeries`, `IsIdealHPolynomial`.
- `hilbertFunction_monomialIdeal`: Lemma 2 for every finite variable set.
- `hilbertFunction_I4`, `hilbertSeries_I4`, `isHPolynomial_I4`, `isHPolynomial_unique`, `h4_tripleDescent`, `counterexample`.
- `idealHF_add_HF`, `finrank_homogeneous_four`, `idealHilbertSeries_I4_mul`, `isIdealHPolynomial_I4`, `isIdealHPolynomial_unique`, `hI_tripleDescent`.
- Main theorems, for every field K :
`conjecture_false (K) : Not (forall n (I : Ideal (MvPolynomial (Fin n) K)), IsQuadraticMonomialIdeal I -> forall h, IsHPolynomial I h -> Not (HasTripleDescent h))`
and `conjecture_false_ideal_reading`, the same with `IsIdealHPolynomial`.

- Axiom audit: `#print axioms` for both main theorems and `counterexample` reports only `propext`, `Classical.choice`, `Quot.sound`. No `sorry`, no `native_decide`.